

Mahipal Thakur

Senior Full Stack Software Engineer | .NET | Azure | Angular | Cloud-Native

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PROFILE

Senior .NET Full-Stack Developer with 6 years building cloud-native enterprise applications on Azure, AWS, and GCP. At Ecolab, architected event-driven microservices across 15+ services, implemented CI/CD pipelines that cut deployment time by 40%, and maintained 90%+ unit test coverage. Hands-on experience with Docker, Kubernetes, GCP, BigQuery, and reinforcement learning (RL) systems. Proficient in Agile/Scrum delivery with a strong culture of code review and test-driven development (TDD).

AREAS OF EXPERTISE

ASP.NET Core • C# • REST APIs • GraphQL / OpenAPI / Swagger • Microservices • gRPC • CQRS • Domain-Driven Design • Microsoft Azure • AWS • GCP • BigQuery • Docker • Kubernetes • Terraform • CI/CD • DevOps • Angular • RxJS • SQL Server • Redis • Entity Framework • JWT Authentication • Azure Functions • Azure Service Bus • xUnit • NUnit • Moq • TDD • Serilog • Application Insights

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer | **Ecolab** | Bangalore, KA | 06/2024 – Present

Project: Cloud-based Enterprise Application Platform

Developing scalable cloud-native enterprise applications using Microsoft Azure, .NET, and Angular. The platform leverages event-driven architecture, REST APIs, and modern frontend frameworks to support high-performance business workflows and secure distributed systems.

- Designed and developed scalable backend services and RESTful APIs (OpenAPI/Swagger) using ASP.NET Core and Microsoft Azure, following Domain-Driven Design (DDD) and CQRS patterns, supporting 500+ enterprise users across 3 geographies.
- Implemented serverless and event-driven architectures using Azure Functions, Azure Service Bus, and Azure Storage Queues, reducing async message processing latency by 35% across 15+ microservices.
- Built and maintained CI/CD pipelines via Azure DevOps, enabling automated builds, integration testing, and zero-downtime deployments — reducing release cycle time by 40%.
- Built secure authentication and authorization using JWT Authentication and RBAC; integrated Redis for distributed caching, improving API response times by 10%.
- Built high-performance Angular applications with reusable components, RxJS-based reactive data flows, lazy loading, and Angular Material, reducing initial load time by 25% across key modules.
- Led code review practices and enforced coding standards; championed TDD with xUnit and Moq, maintaining 90%+ unit and integration test coverage across 15+ microservices.
- Enhanced system reliability and observability using structured logging (Serilog), retry strategies, and Application Insights monitoring; conducted Agile/Scrum ceremonies as an active contributor.
- Deployed and managed containerized services using Docker and Kubernetes on Azure App Service and Azure Container Registry, following DevOps best practices.
- Developed end-to-end API endpoints implementing business logic, request validation, and integration with multiple cloud services, documented via OpenAPI/Swagger.
- Collaborated with cross-functional teams of 8+ frontend developers, QA engineers, and DevOps in Agile/Scrum sprints to deliver reliable cloud-native applications.

Technology:

.NET / Backend: ASP.NET Core, C#, RESTful APIs, OpenAPI/Swagger, gRPC, CQRS, DDD, Microservices, Entity Framework, LINQ, ADO.NET, SQL Server, Stored Procedures, Dependency Injection, Middleware, JWT Auth, RBAC, Redis, Exception Handling, xUnit, NUnit, Moq, TDD

Frontend: Angular, RxJS, Angular Material, TypeScript, Reusable Component Libraries

Cloud (Azure): Azure Functions, Azure Service Bus, Azure Storage Queues, Azure DevOps, Azure App Service, Azure Container Registry, Application Insights, CI/CD

Cloud (AWS): EC2, VPC, Public/Private Subnets, Internet Gateway, NAT Gateway, Elastic IP, AWS Lambda, EventBridge, CloudWatch

Containers & DevOps: Docker, Kubernetes, Terraform, CI/CD, Azure DevOps

Observability: Serilog, Application Insights, structured logging, retry strategies

Full-Stack Software Engineer | [HappiestMinds](#) | Bangalore, KA | 11/2021 – 06/2024

Project: Ascend Learning (Learning Management System)

- Led end-to-end design, development, and rigorous testing (TDD) of LMS modules serving 200+ concurrent users; conducted regular code reviews to enforce quality and maintainability standards.
- Architected and optimized a robust business logic layer using ASP.NET Core and Angular, streamlining data processing and improving system responsiveness by 20% within an Agile/Scrum team.
- Developed and implemented SQL scripts and stored procedures to automate data manipulation, using temporary tables to reduce data processing time by 1.5%.
- Documented APIs using OpenAPI/Swagger and participated in daily Scrum meetings with onshore and offshore teams across 2 time zones to synchronize project progress.

Technology: ASP.NET MVC, ASP.NET Core Web API, Angular 8+, SQL Server, C#, VB, N-tier & Microservices architecture, xUnit, Swagger/OpenAPI

Software Engineer | iConcept | Hyderabad | 04/2021 – 11/2021

Project: AgriTech — solutions for crop, field, channel, and market challenges

- Optimized stored procedures, AJAX data binding, and SQL scripts within MS SQL Server, improving query performance by 20%.
- Developed Web APIs using .NET Core with advanced data sorting, filtering, and structured logging using Serilog; integrated Kafka for event streaming in a CI/CD-enabled pipeline.

Technology: ASP.NET MVC, JavaScript, jQuery, HTML, CSS, Bootstrap, AJAX, LINQ, Entity Framework, Stored Procedures, Serilog, Kafka

PROJECTS

[PProbe](#) — RL-based Code Review Agent | Meta PyTorch OpenEnv Hackathon | | 2026

Built an AI-powered code review agent using reinforcement learning (GRPO) to detect security vulnerabilities (SQL injection, hardcoded credentials) in pull requests. Trained Qwen2.5-1.5B-Instruct on Kaggle/Google Colab; qualified for Final Round.

- Fine-tuned Qwen2.5-1.5B-Instruct using GRPO (Group Relative Policy Optimization) on a custom security vulnerability dataset covering SQL injection, hardcoded credentials, and insecure patterns.
- Built an OpenEnv-compliant environment with typed Observation, Action, and Reward Pydantic models; implemented programmatic graders scoring 0.0–1.0 across easy/medium/hard tasks.
- Ran multiple successful training iterations on Kaggle and Google Colab; achieved qualification for Hackathon Round 1 out of hundreds of global submissions.

Tech Stack: Python, PyTorch, Reinforcement Learning (GRPO), OpenEnv, Qwen2.5-1.5B-Instruct, Pydantic, Kaggle, Google Colab

SKILLS

- **Frontend:** Angular, TypeScript, RxJS, Angular Material
- **Backend:** ASP.NET Core, C#, REST APIs, gRPC, GraphQL, OpenAPI/Swagger, CQRS, DDD, Microservices
- **Cloud (Azure):** Azure Functions, Azure Service Bus, Azure DevOps, Azure App Service, CI/CD, Application Insights
- **Cloud (AWS):** EC2, VPC, AWS Lambda, EventBridge, CloudWatch
- **Cloud (GCP):** GCP, BigQuery, ML APIs
- **Data & ORM:** SQL Server, Entity Framework, ADO.NET, Redis, Stored Procedures
- **Auth & Security:** JWT Authentication, RBAC, Middleware Pipeline
- **Testing:** xUnit, NUnit, Moq, Jasmine, TDD, Integration Testing, 90%+ coverage
- **Containers & DevOps:** Docker, Kubernetes, Terraform, CI/CD, Azure DevOps, GitLab
- **Observability:** Serilog, Application Insights, structured logging
- **Tools:** JIRA, GitHub Copilot, VS Code, Visual Studio
- **AI / ML:** Reinforcement Learning (GRPO), PyTorch, OpenEnv, BigQuery ML

ONLINE COURSES & CERTIFICATIONS

- Mastering Angular Certification Training (Aug 2021 – Mar 2022) — DotNetTricks | ID: J3WB170322
- REST API with ASP.NET Core Web API — Udemy (Jan 2021 – Jul 2021)
- Trained on AZ-900 | AZ-204 | AZ-104 | AZ-305 (Certified by SimpliLearn) — Target: Q3 2026
- AWS Certified Cloud Practitioner (in progress) — Target: Q3 2026
- **Cloud Architect Masters Program — SimpliLearn** | 2026 – Present | Focus: AWS, Azure, GCP, Infrastructure Design, CI/CD
- **Modern Software and AI Engineering Program — Scaler Academy** | 2026 – Present | Focus: DSA, System Design, Distributed System
- **Google Cloud Gen AI Academy APAC 2026 — (Google Cloud & Hack2skill)** | 2026 – Present

EDUCATION

B.Tech in Computer Science — MAKAUT, WB, Kolkata, West Bengal | 2015 – 2019